

Week 1 Weekend Assignment

"Network Detective Challenge"

Due: End of Week 1 (before Day 5 test) | Time estimate: 3-4 hours across Saturday + Sunday

What you need: Your own laptop/PC, Wireshark (free), internet connection, this document

The Big Picture

Over the past 4 days you learned:

- Day 1: What cybersecurity is — CIA Triad, AAA, threats vs vulnerabilities vs risks
- Day 2: Networking basics — OSI model, TCP/IP, IP addressing, ports, protocols
- Day 3: Common protocols — HTTP/HTTPS, FTP/SFTP, SSH, SMTP, DNS, ICMP + subnetting
- Day 4: Linux & Windows security — permissions, users, hardening, event logs

Now it is time to combine all of that into one real-world mini-project.

This assignment has 3 parts. Do all 3. They build on each other.

Part 1 — Capture & Analyze HTTP Traffic (Day 2 + Day 3 skills)

What to do:

1. Install Wireshark (if not already): <https://www.wireshark.org/download.html>
2. Open Wireshark and start capturing on your active network interface (Wi-Fi or Ethernet).
3. While capturing, open a browser and visit 3 different websites:
 - One that uses HTTP (try <http://neverssl.com> — it deliberately uses HTTP)
 - One that uses HTTPS (try <https://www.google.com>)
 - One of your choice (any site you normally use)
4. Stop the capture after about 60 seconds of browsing.
5. Apply the display filter `http` in Wireshark — this shows only HTTP traffic.
6. Also try the filter `dns` to see DNS queries.
7. Also try the filter `tcp.port == 443` to see HTTPS traffic.

What to write about (answer in your own words):

For each of the 3 websites, answer:

Question	Your answer
What website did you visit?	www.google.com
HTTP or HTTPS? How can you tell from the capture?	https, all the information was encrypted
What IP address did your computer connect to?	By using <code>nslookup</code> command we can tell, it connected to google's ip address which is 142.250.146.100
What was the destination port? (80 or 443?)	Port 443
For HTTP sites: what information is visible in the packet?	I visited an encrypted site
For HTTPS sites: what information is visible vs hidden?	Number of frames, bytes, protocol used, synchronization and acknowledgements, these were visible. Information like application data of the website were completely hidden
What DNS queries did you see before the connection?	A standard query was sent from my pc ip address to router, the router fetches the domain ip address from isp and sends it back to my pc

Key concepts to connect:

- CIA Triad: When you visited the HTTP site, which part of CIA was violated? (Hint: no encryption = no Confidentiality)

Answer: The data of the website transfers in plain text, without any encryption which leads to security risk, in CIA triad, the confidentiality of the person's data is breached

- AAA: If the HTTP site had a login page, which AAA function would be exposed? (Hint: Authentication credentials)

in clear text)

Answer: The authentication fails here, if the login page has no encryption while sending data to database, our information is unsafe because it is hiding in plain sight.

• Subnetting: What is the IP range of your home network? (Check with ipconfig or ifconfig) What subnet mask are you using?

Answer: 103.159.249.139

Where to get help /sources:

- Wireshark official docs: <https://www.wireshark.org/docs/>
- Wireshark 101 tutorial: <https://www.youtube.com/watch?v=TkCSr30UojM>
- How to filter HTTP traffic: In Wireshark, type http in the display filter bar

Part 2 — Subnetting Practice (Day 3 skills)

Using what you learned about subnetting in Day 3, solve these problems. Show your working.

Problem 1

Your home router gives you the network 192.168.1.0/24.

- How many total host addresses are available in this network?

Answer: 24 bits are part of network, we have remaining of 8 bits $(32-24) = 2^8=256$ total addresses

- If you split it into 4 equal subnets, what is the new subnet mask in CIDR notation?

Answer: If split into four ways, $2^2=4$, two bits are added to host address which makes it /26 instead of /24

- What are the network addresses of all 4 subnets?

192.168.1.0

192.168.1.64

192.168.1.128

192.168.1.192

- What is the broadcast address of the third subnet?

Answer: Broadcast address is the last address of the subnet which is 192.168.1.191

- How many usable hosts per subnet after splitting?

Answer: $64-2=62$ addresses, the first and last addresses are used for network and broadcast

Problem 2

You are a network admin for a small office. You have these departments:

- Sales: needs 30 hosts
- Engineering: needs 50 hosts
- HR: needs 10 hosts
- Servers: needs 10 hosts

You are given the network 10.0.0.0/24.

- Design a subnetting scheme that gives each department enough addresses.
- For each department, write: network address, subnet mask (CIDR), usable host range, broadcast address.
- Which department gets the largest subnet? Why?

Engineering — /26 (64 addresses)

- **Network:** 10.0.0.0/26
- **Usable range:** 10.0.0.1 → 10.0.0.62
- **Broadcast:** 10.0.0.63

Sales — /27 (32 addresses)

- **Network:** 10.0.0.64/27
- **Usable range:** 10.0.0.65 → 10.0.0.94
- **Broadcast:** 10.0.0.95

HR — /28 (16 addresses)

- **Network:** 10.0.0.96/28
- **Usable range:** 10.0.0.97 → 10.0.0.110

- **Broadcast:** 10.0.0.111

Servers — /28 (16 addresses)

- **Network:** 10.0.0.112/28
- **Usable range:** 10.0.0.113 → 10.0.0.126
- **Broadcast:** 10.0.0.127

The department that gets the largest subnet is engineering as they need 50 different hosts.

Problem 3 — Connect to Security

- Why would a security team want to put the Servers department on a different subnet from Sales? (Answer using CIA + AAA concepts)

Answer: The security team wants to put each department in a different subnet because, it creates maintainability and a person of one department cannot access to another department which becomes a security risk, to maintain accountability and monitor activity of the employees different subnets are used.

- If an attacker compromises a Sales PC, can they directly access a Server on a different subnet? Explain why or why not.

Answer: thankfully a compromised sales pc poses no threat to the finance department as they are in a different subnets which can be difficult to access other subnets therefore subnetting is crucial

- What device controls traffic between subnets? (Hint: think about Day 2 — routers/firewalls)

A router is used to control and direct traffic, but if need to security for these firewalls are used

Part 3 — Connect Everything: CIA + AAA Across All 4 Days

This is the thinking part. Write short paragraphs (3-5 sentences each). No copy-paste — use your own words.

Question 1 — CIA Triad in Action

Pick one of the protocols you studied (HTTP, FTP, SMTP, DNS, SSH, or ICMP). For that protocol:

- How does it support one part of the CIA Triad?
- How does it fail to protect one part of the CIA Triad?
- What would a security professional do to fix that weakness?

Answer: I choose the DNS protocol, The part which is supported by the CIA triad is availability, Ensures any authorized user with a router gets to use the DNS protocol, which is used to translate, domain names into ip addresses.

DNS protocol protects all parts of CIA, confidentiality, integrity, availability.

Question 2 — AAA Framework in Action

Think about logging into a website (like Gmail or any site you use).

- Authentication: How does the site verify you are who you say you are?
- Authorization: After you log in, what are you allowed to do? What are you NOT allowed to do?
- Accounting: What records might the site keep about your login? Why?

Now answer: What happens if an attacker captures your HTTP traffic from Part 1? Which part of AAA breaks first? Why?

I chose the AWS website, for this question, Authentication: AWS uses multi factor authentication like password and username and codes that are changed every thirty seconds or biometrics, which finds that I am who I am saying.

Authorization: if I login from my root account, it gives me the full dashboard of the aws, if login from other credentials, only information allotted to me can be used.

Accounting: my every click, timestamps of login etc are saved.

Question 3 — The Full Attack Story

Imagine this scenario:

An employee at a company uses an unencrypted HTTP connection to log into the internal HR portal from a coffee shop. The company's network is poorly segmented — all departments are on the same subnet 192.168.1.0/24.

Using everything you learned this week, write a short attack story:

1. What can the attacker on the coffee shop Wi-Fi see? (Day 3 — HTTP, packet capture)
2. What can the attacker do with those credentials? (Day 1 — Authentication broken, CIA violated)
3. Once inside the company network, what can the attacker reach? (Day 3 — subnetting, no segmentation)
4. How could the company have prevented this? (Day 4 — hardening, Day 3 — segmentation, Day 1 — controls)

An employee sitting in a coffee shop doing her work logs into the internal HR portal, the attacker at the coffee shop can see her credentials which can be seen easily using Wireshark, the internal data password username are all available to the hacker, there was no multi factor authentication or credential rotation, a single password and username transferred in plain text, which is held by the attacker, every part of CIA is violated, once the attacker reaches the company network he can see there's no subnetting to each department, a single subnet is shared across the company, where you can easily access others data. The company could've prevented this by enabling a few steps which are: using a secure transfer protocol like https instead of http, rotation of credentials, multi factor authentication, subnetting each department so it is an extra layer of protection to the company. With these simple changes the company could've prevented the hacking of the site.

Question 4 — Linux/Windows Connection

- On Day 4 you learned about file permissions (Linux) and Event Viewer (Windows).
- How does the principle of least privilege (Day 4) connect to the AAA Authorization function (Day 1)?
Answer: Authorization this is where the least privilege happens, the user gets access to things only needed and nothing more, Accounting: Every move is logged so when suspicious activity arises you can look at it.
- If an attacker gets a shell on a Linux machine, what permission-related files would they look for?
The first thing an attacker does is figure out who they are and what they can reach. Example: commands like `passwd`, `shadow`, `sudoers` are used to check for permissions and passwords.
- On Windows, where would you look in Event Viewer to spot a brute-force login attempt?
Event viewer => window logs => security, the event id determines how the login attempt went, brute force attempt can be easily spotted, when there are multiple failed attempts followed by a successful login

How to Submit

Create a single document (text file, markdown, or PDF) with:

1. Part 1: Your traffic analysis table + answers + at least 2 screenshots from Wireshark
2. Part 2: All subnetting answers with working shown
3. Part 3: All 4 written answers in your own words

Name it: Week1_Weekend_Assignment_YourName.md (or .txt or .pdf)

Grading Rubric (self-grade before submitting)

Criteria	Points	Your score
Part 1: Traffic captured and analyzed correctly	/25	20
Part 1: CIA + AAA connections made	/10	10
Part 2: Subnetting calculations correct	/25	25
Part 2: Security reasoning about subnets	/10	10
Part 3: Own words, no copy-paste	/15	15

Part 3: Connections across all 4 days are accurate	/15	12
Total	/100	92

Tips for Doing This Well

- Do Part 1 first. It is the most hands-on and will make the theory in Parts 2 and 3 feel real.
- Use your notes from all 4 days. This assignment is designed to make you go back and review everything.

- If you get stuck on subnetting, re-read the Day 3 notes. The 192.168.1.0/24 example is your starting point.
- Screenshots matter. In cybersecurity, evidence matters. Show what you captured.
- Write like you are explaining to a friend. If you cannot explain it simply, you do not understand it yet.